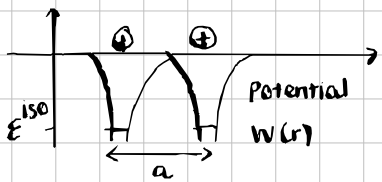


Electrons in Crystal

In Hydrogen Atom

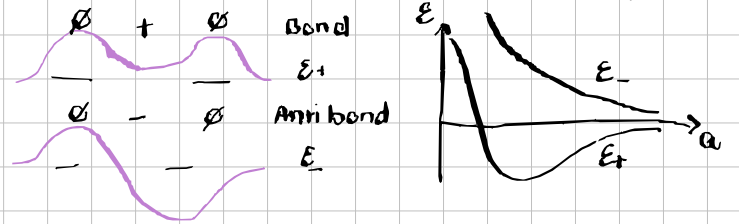


It has the Hamiltonian of

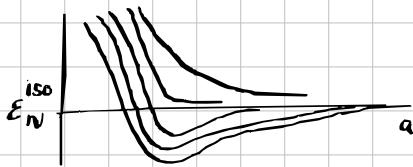
$$\hat{H} = \frac{\hat{p}^2}{2m} + W(\vec{r}) = -\frac{\hbar^2}{2m} \nabla^2 + W(\vec{r})$$

with ground state ϕ

With two atoms, one electron can be in a superposition



The superposition phase for this one electron creates different energy levels



At certain a , degeneracy is lifted at most N
But at $a \rightarrow \infty$, the energy is degenerate again.



Even though, as $a \rightarrow \infty$ it seems like the electron should be localized for only one atom. That is correct except that think of it as you start with a superposition and separate or you start with a "single" atom. This thinking gives

localized or "localized"

Consider this electron as free particle is the eigenstate

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 \cdot \text{A plane wave } \psi_{\vec{k}}(\vec{r}) \sim e^{i\vec{k} \cdot \vec{r}}$$

$$\vec{p} = \hbar \vec{k} \quad E(\vec{k}) = \frac{\hbar^2 |\vec{k}|^2}{2m}$$

$$\vec{p} \rightarrow \vec{k}$$



Note that as $\vec{k} \uparrow$, it does not repeat. Lattice allows the repetition

But in a lattice, there is also a potential:

$$\left(-\frac{\hbar^2 \nabla^2}{2m} + W(\vec{r}) \right) \phi(\vec{r}) = E \phi(\vec{r}) \quad \text{with } W(\vec{r} + \vec{R}_n) = W(\vec{r})$$

The solution becomes:

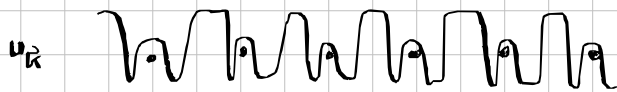
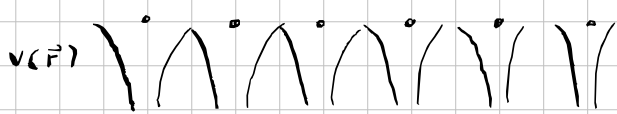
$$\psi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}(\vec{r})$$

$$u_{\vec{k}}(\vec{r} + \vec{R}_n) = u_{\vec{k}}(\vec{r}) \quad \begin{matrix} \nearrow \text{make } G \text{ discrete} \\ \searrow \text{band index / Energy level} \end{matrix}$$

The Bloch theorem says

$$\psi_{\vec{k}}(\vec{r} + \vec{R}) = \psi_{\vec{k}}(\vec{r}) e^{i\vec{k} \cdot \vec{R}}$$

where $\vec{k} \cdot \vec{R}$ can be anything!



Thus for a general structure:

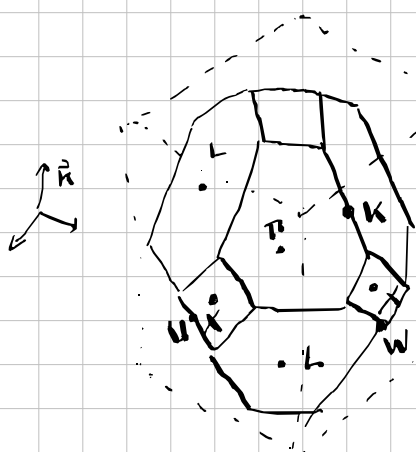
$$\left(-\frac{\hbar^2 \nabla^2}{2m} + W(\vec{r}) \right) \phi_{\vec{k}}(\vec{r}) = E_{\vec{k}} \phi_{\vec{k}}(\vec{r})$$

W : lattice potential $E_{\vec{k}}$: energy band structure

$$\phi_{\vec{k} + \vec{G}}(\vec{r}) = \phi_{\vec{k}}(\vec{r}) \quad E_{\vec{k} + \vec{G}} = E_{\vec{k}}$$

Even when we look at R_n away from last position or the electron wave vector is increased by G . The energy is the same

Observe the reciprocal lattice of an FCC crystal (BCC reciprocal), We start with $\Gamma = \vec{k} = 0$



X is center of square face

L is center of hexagon face

W is vertex

K midpoint of edge between two hexagon

U midpoint of a square and hexagon

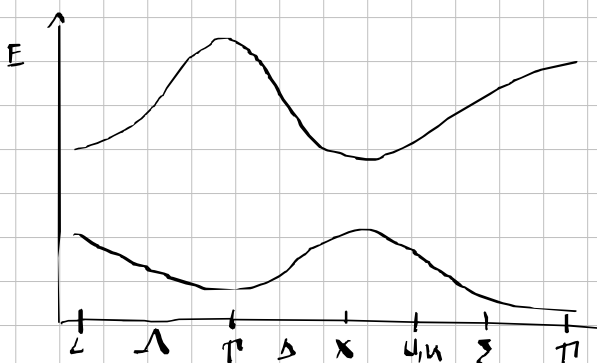
Δ : $\Gamma \rightarrow X$

Λ : $\Gamma \rightarrow L$

G : $L \rightarrow W$

Σ : $X \rightarrow U$

When a band structure plot has a horizontal axis such as $\Gamma \rightarrow X \rightarrow W \rightarrow K \rightarrow L$, it means we calculate the energy along these directions. k in the x-axis is not magnitude



Caveats: The Bloch theorem allows $\vec{k} \cdot \vec{R}$ to be any phase. This says that any electron with any $\vec{k}$ is allowed in the crystal. However a $\vec{k} + \vec{G}$ is reproducing the same wave function. G is discrete: allowed shift while $\vec{k}$ is continuous in an infinite crystal.

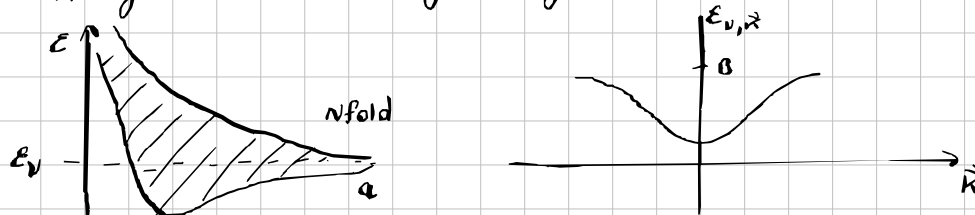
Tight Binding Model

Consider putting atoms closer such as there is an overlapping with wave functions



The overlapping lifts up Nfold degeneracy

This creates a continuous band



We will see how this energy band arrives, consider the Schrödinger

$$\left\{ -\frac{\hbar^2}{2m} \nabla^2 + W(\vec{r}) \right\} \phi_{v\vec{k}}(\vec{r}) = E_{v\vec{k}} \phi_{v\vec{k}}(\vec{r})$$

In one dimension model $\vec{r} \rightarrow x$

$$W(x) = \sum_{\ell} w_{ion}(x - R_{\ell}) \quad \text{with } R_{\ell} = \ell \cdot a \quad N \rightarrow \infty$$

The solution for each atom is

$$H^0 \phi_v(x) = E_v^0 \phi_v(x) \quad \text{with } H^0 = -\frac{\hbar^2}{2m} \nabla^2 + w_{ion}(x)$$

Construct

$$\phi_{v\vec{k}}(x) = \frac{1}{\sqrt{N}} \sum_{\ell} e^{i\vec{k} \cdot R_{\ell}} \phi_v(x - R_{\ell}) \quad E_{v\vec{k}} = \frac{\int dx \phi_{v\vec{k}}^*(x) H \phi_{v\vec{k}}(x)}{\int dx \phi_{v\vec{k}}^*(x) \phi_{v\vec{k}}(x)}$$

This is a superposition of an electron with a phase $e^{i\vec{k} \cdot R_{\ell}}$. This phase changes from one site to other.

$$E_{v\vec{k}} = \frac{1}{N} \sum_{nm} e^{-i\vec{k} \cdot (R_n - R_m)} \int dx \phi_v^*(x - R_n) \{ E_v^0 + \Delta V_{L,m}(x) \} \phi_v(x - R_m) \quad \Delta V_{L,m} = \sum_{\ell \neq m} w_{ion}(x - R_{\ell})$$

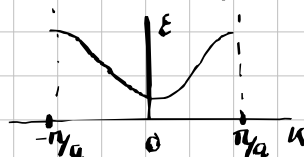
Where other nearby potentials near m contribute the perturbation

Assume small overlap

$$\int dx \phi_v^*(x - R_n) \Delta V_{L,m}(x) \phi_v(x - R_m) \approx \begin{matrix} \text{First order} \\ \downarrow \\ V_0 \delta_{nm} + V_1 \delta_{n,m+1} + V_{-1}^* \delta_{n,m-1} \end{matrix}$$

$$E_{v\vec{k}} = E_v^0 + V_0 + V_1 \frac{1}{N} \sum_{nm} \left(e^{-i\vec{k} \cdot (R_{m+1} - R_m)} \delta_{n,m+1} + e^{-i\vec{k} \cdot (R_{m-1} - R_m)} \delta_{n,m-1} \right) \quad R_{m+1} - R_m = \pm a$$

$$= E_v^0 + V_0 + 2V_1 \cos(ka)$$



This perturbation theory predicts a cosine structure for the band

An electron near the bottom of a conduction band behaves approximately like a free electron, but with an effective mass

At $\vec{r} = \vec{k} = 0$, expand Taylor series

$$E_{c,\vec{k}} \approx E_{c,0} + \frac{1}{2} \left. \frac{\partial^2 E_{c,\vec{k}}}{\partial k^2} \right|_{\vec{k}=0} k^2 = E_{c,0} + \underbrace{\frac{\hbar^2 k^2}{2m_c}}_{\text{free energy}} \quad \text{At minimum } \vec{r} \quad \frac{\partial E}{\partial \vec{k}} = 0$$

$E_{c\vec{k}}$ is dispersion of free particle with effective mass m_c

Periodic Boundary Condition

Until now, the volume is essentially infinite. But, in reality the volume $V = L^3$ is finite.



If we make $\psi(x+L) = \psi(x)$ in all direction. This is mathematically the same as infinite crystal. That is

$$e^{i k_x (x+L)} = e^{i k_x x} \Rightarrow k_x L = 2\pi$$

Finite crystal makes the wavelength finite.

$$k_x = m \frac{2\pi}{L} \quad \Delta k_x = \frac{2\pi}{L} \quad \Delta^3 k = \Delta k_x \Delta k_y \Delta k_z = \left(\frac{2\pi}{L}\right)^3$$

As $L \rightarrow \infty$, this k is continuous.

Using

$$\sum_{\vec{k}} = \frac{1}{\Delta^3 k} \sum \Delta^3 k \rightarrow \left(\frac{L}{2\pi}\right)^3 \int d^3 k$$

Density of states can be evaluated

$$2 \frac{1}{V} \sum_{\vec{k}} f(E_{\vec{k}})$$

As $V \rightarrow \infty$, $\vec{k}$ is continuous,

We ask how many states have energy near E rather than what are all individual $\vec{k}$ states

Example is the density of electrons in a band

$$n_e = \frac{N_e}{V} = 2 \frac{1}{V} \sum_{\vec{k}} f(E_{\vec{k}}) \rightarrow \int dE \tilde{g}_V(E) f(E)$$

$$g_V(E) = \frac{\text{number of available states } \vec{k} \text{ around } E \text{ in band } V}{(\text{unit volume}) (\text{unit energy})}$$

Formally, one can write as how many $\vec{k}$ contain same energy.

$$g_V(E) = \frac{2}{V} \sum_{\vec{k}} \delta(E - E_{\vec{k}}) \Rightarrow \int dE g_V(E) f(E) = \frac{2}{V} \sum_{\vec{k}} f(E_{\vec{k}})$$

Since tight binding model gives

$$E_{\vec{k}} = E_{v0} + \frac{\hbar^2 k^2}{2m_v} \Rightarrow k = \frac{\sqrt{2m_v(E - E_{v0})}}{\hbar}$$

The amount of k -states lie in thin spherical cell is

$$dN \propto 4\pi k^2 dk$$

$$g_V(E) \propto k^2 \frac{dk}{dE}$$

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m_v} \quad \frac{dk}{dE} = \frac{m_v}{\hbar^2 k} \Rightarrow k^2 \frac{dk}{dE} \propto m_v^{3/2} \sqrt{E - E_{v0}}$$

$$g_V(E) = \frac{1}{2\pi^2} \left(\frac{2m_v}{\hbar^2} \right)^{3/2} \sqrt{E - E_{v0}} \quad \Theta(E - E_{v0})$$



Electron Distribution

Consider how likely each electron state in a band can be occupied

$$f_v(\vec{k}) = \langle a_{v\vec{k}}^\dagger a_{v\vec{k}} \rangle = \frac{1}{e^{\beta(E_v(\vec{k}) - \mu)} + 1} \Rightarrow \text{Lower energy allows better occupation}$$

$\beta = 1/(k_B T)$

v : band index $\vec{k}$ is crystal vector and $E_v(\vec{k})$ is the energy of that state μ is chemical potential
 f indicates the average number of particles in state $v, \vec{k}$

The chemical potential is chosen such that

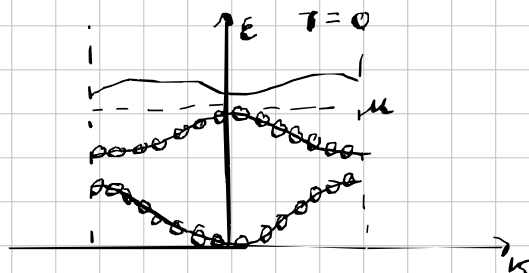
$$N_e = \sum_{v, \vec{k}} f_v(\vec{k})$$

Particle density

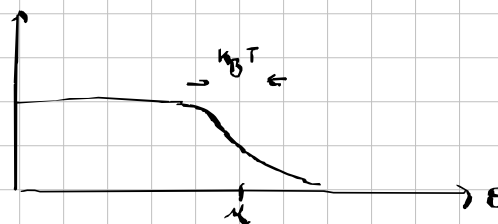
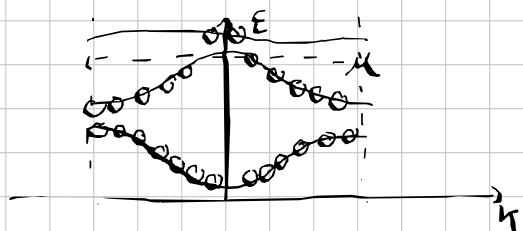
$$n = \frac{N}{V}$$

At limit $T \rightarrow 0$, if $E < \mu$, $e^{\beta(E - \mu)} \rightarrow 0$ $f_v(\vec{k}) \rightarrow 1$
 if $E > \mu$, $e^{\beta(E - \mu)} \rightarrow \infty$ $f_v(\vec{k}) \rightarrow 0$

At zero temperature, there is no air particle kicking the electron so that it can only occupy states below μ
 When temperature increases, some particles have more energy to occupy above μ



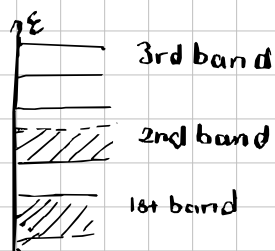
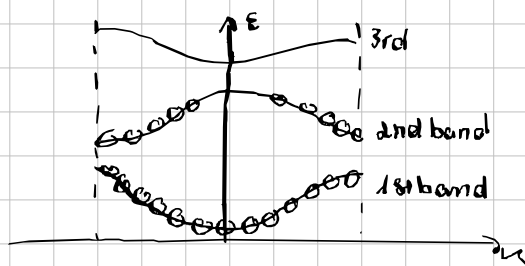
$T > 0$: Fermi edge



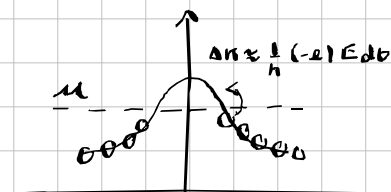
When T is large enough $\beta \sim 0$, this is Maxwell distribution $f(E) \rightarrow e^{-\beta(E - \mu)}$

Classification of Solid

At Fermi distribution, $T \sim 0$ the typical scale for E is eV since $k_B T \sim meV$ even at $300K$

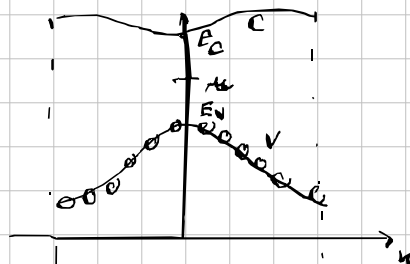


If μ is within a band, it is metal



An applied static E field can easily accelerate electrons in the band producing electrical current

If μ is between bands, it is insulator / semiconductor



The electrons in valence band create no current due to symmetry, but yet has no energy to excite to upper band.

Summary:

